

ANCHORAGE INTL, AK, USA

WMO: 702730

Lat: 61.1567N

Lon: 149.9864W

Elev: 144

StdP: 14.62

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 26451

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.4	-2.8	-15.0	2.4	-3.1	-11.6	3.0	0.3	24.7	30.8	21.0	28.7	2.8	10	0.687

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.8	72.5	59.7	69.2	57.9	66.7	56.8	61.2	70.3	59.4	67.1	58.1	64.8	7.4	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
57.0	69.7	64.1	55.6	66.2	62.5	54.4	63.3	61.3	27.1	70.4	26.0	67.4	25.1	64.7	70.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.9	18.7	16.7	-11.4	77.0	5.9	4.2	-15.7	80.0	-19.1	82.5	-22.5	84.8	-26.8	87.9		
(5)			-11.9	63.6	5.8	2.5	-16.0	65.4	-19.4	66.9	-22.7	68.3	-26.9	70.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	38.0	17.7	22.0	26.2	37.8	48.3	56.0	59.6	57.6	49.7	36.8	23.9	19.5		
	DBStd	17.06	12.13	10.03	8.56	6.09	5.10	4.21	3.70	3.64	5.02	7.95	10.06	10.32		
	HDD50	5187	1001	783	737	367	93	3	0	1	64	410	784	944		
	HDD65	9859	1466	1203	1202	817	519	273	173	231	458	874	1234	1409		
	CDD50	812	0	0	0	0	40	182	297	237	55	1	0	0		
	CDD65	10	0	0	0	0	0	2	6	2	0	0	0	0		
	CDH74	48	0	0	0	0	0	16	25	7	0	0	0	0		
	CDH80	4	0	0	0	0	0	1	3	0	0	0	0	0		
	WSAvg	6.6	5.9	6.0	6.8	7.0	8.1	7.8	6.9	6.3	6.3	6.2	6.2	6.0		
	PrecAvg	16.4	0.8	0.8	0.7	0.5	0.7	1.1	1.8	2.8	3.0	1.9	1.2	1.1		
(15) Precipitation	PrecMax	27.6	2.7	2.4	2.8	1.9	1.9	3.4	4.5	9.8	7.7	4.3	2.8	2.7		
	PrecMin	8.1	0.0	0.1	0.0	0.0	0.0	0.2	0.2	0.3	0.7	0.4	0.0	0.1		
	PrecStd	3.5	0.6	0.5	0.5	0.5	0.4	0.7	1.0	1.8	1.7	0.9	0.8	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.1	42.4	46.5	57.5	70.5	76.5	77.1	74.8	65.0	54.5	46.0	43.8		
		MCWB	37.6	37.2	39.3	45.6	53.1	60.5	62.6	61.5	54.8	47.6	40.7	38.1		
	2%	DB	39.6	39.5	42.4	52.8	65.2	71.1	72.8	70.3	61.5	51.2	42.3	39.9		
		MCWB	35.5	35.5	36.7	42.5	51.1	57.6	60.9	59.6	53.2	45.3	38.9	36.3		
	5%	DB	36.1	36.7	39.7	49.9	61.0	67.5	69.4	67.1	59.2	49.1	39.3	35.6		
		MCWB	33.3	33.7	34.7	40.9	49.0	56.0	58.9	57.9	52.0	44.1	36.5	33.1		
	10%	DB	32.6	34.0	37.3	47.3	57.6	64.4	66.9	64.8	57.1	47.1	36.5	32.7		
		MCWB	30.7	31.7	32.9	39.7	47.4	54.4	57.8	56.7	51.1	42.8	34.1	30.9		
	0.4%	WB	38.1	37.5	40.2	46.8	54.7	62.1	64.0	63.0	56.0	49.3	41.3	38.5		
		MCDB	42.2	40.9	45.3	56.4	68.2	73.9	74.6	72.3	62.2	53.0	44.5	42.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.8	35.7	37.3	43.5	51.8	59.0	61.6	60.7	54.4	46.5	39.3	36.1		
		MCDB	39.3	38.8	41.6	50.9	62.8	68.9	71.3	68.5	59.3	49.9	42.1	39.6		
	5%	WB	33.3	34.0	35.3	41.6	50.0	57.0	59.9	59.1	53.2	45.0	36.9	33.2		
		MCDB	35.6	36.3	39.0	48.7	59.5	65.5	67.9	65.6	57.5	48.4	39.1	35.0		
	10%	WB	30.9	32.0	33.4	40.2	48.2	55.2	58.5	57.6	52.0	43.4	34.3	31.0		
		MCDB	32.7	34.1	36.7	46.7	56.6	63.1	65.5	63.0	55.9	46.6	36.1	32.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.8	10.4	12.8	13.8	14.7	13.6	11.8	11.9	11.5	10.0	9.3	9.7		
		MCDBR	11.8	10.8	12.6	17.4	21.1	19.6	17.1	16.5	14.3	10.9	9.8	12.0		
		MCWBR	8.9	8.1	8.4	10.2	10.9	9.7	8.6	8.3	8.1	6.7	7.4	9.4		
	5% WB	MCDBR	11.2	10.4	11.8	16.0	18.4	17.8	15.5	14.5	12.4	9.4	9.3	11.1		
		MCWBR	8.6	8.0	8.1	9.7	10.0	9.4	8.1	7.8	7.4	6.0	7.1	9.0		
(40) Clear-Sky Solar Irradiance	taub		0.234	0.287	0.305	0.346	0.369	0.358	0.371	0.355	0.318	0.298	0.243	N/A		
	taud		2.131	2.235	2.273	2.260	2.320	2.433	2.421	2.471	2.564	2.482	2.135	N/A		
	Ebn at Noon		161	220	259	268	270	276	268	264	258	221	151	N/A		
	Edh at Noon		12	21	28	35	36	32	32	28	22	16	11	N/A		
(44) All-Sky Solar Radiation	RadAvg		91	308	821	1355	1636	1689	1497	1185	772	360	132	47		
	RadStd		13	45	90	121	151	110	135	132	108	43	21	6		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+1.40	+3.22	+2.92	N/A	N/A	N/A	-439	-544	N/A
(47) Regional (25 neighbors)		N/A	+3.80	+3.22	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page